

CST 243-3 Rapid Application Development

Lab Sheet 01

Lab Sheet Title: Java practice questions

1. Create a class called “Rectangle”, allowing you to build a rectangle with length and width attributes.

Create a method called “perimeter” to calculate the perimeter of the rectangle and “area” method to calculate the area of the rectangle.

Create a method called “display” which display the length, width, perimeter and area of the Rectangle. Use appropriate data types and return types for the variables and methods.

Write test class to instantiate the Rectangle class.

2. Write a java program to implement the following scenario.
 - a. Define a class called ‘Employee’. Each employee is represented by Employee No, name, date of birth, salary, and title attributes. Represent all attributes as class-level variables in the Employee class. Add default, empty constructor. Then add another overloaded constructor. Define a method called ‘displayData’ to display all the employee information.
 - b. Define an interface called ‘Employee’. This interface contains the following methods.
 - i. void displayData()
 - ii. double calculateSalary()
 - iii. double calculateBonus(int a)
 - c. Derive two concrete classes, given in the following table, from the interface defined in Question 2 part (b). Implement the business logic given below.

Class Name	Attributes
PermenantEmployee	Service period Bonus Total Salary
TemporaryEmployee	Number of working days per month Bonus Total Salary

Assume that Permenant Employee is paid Rs 50000/= basic salary and Temporary Employee is paid 25000/= basic salary.

To calculate the monthly bonus and salary for Permanent Employees use the following scenario

- Bonus = 25% of the salary for employees having more than 20 years of service
- Bonus = 10% of the salary for employees having between 10 to 20 years of service
- Bonus = 5% of the salary for employees having between 5 to 10 years of service
- No bonus will be added to the salary for employees having less than 5 years of service.

To calculate the monthly bonus and salary for Temporary Employees use the following scenario

- Bonus = 5% of the salary for employees who exceed no of working days more than or equal to 25

Following formula calculates the monthly net salary of both Permanent Employees and Temporary Employees.

$$\text{Total Salary} = \text{Basic Salaty} + \text{Bonus}$$

2. Represent the following two matrices using multidimensional arrays in java.

$$\mathbf{A} = \begin{bmatrix} 2 & 0 & 2 \\ 2 & 2 & 0 \end{bmatrix}$$

$$\mathbf{B} = \begin{bmatrix} 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$$

- a. Print the values of matrix A using loops.
- b. Get the summation of matrix A and matrix B and store the result in matrix C. print the result.

3.

- a. The given grades determine the GPA values as follows:

A=4.00

B=3.00

C=2.00

F=0.00

A system reads the grade and displays the GPA accordingly. Implement the above system using java

- b. Write a program in Java to find the largest value of a given array of 100 integers.
- c. The Following categorization is required to be done:

If the given integer is greater than 75, then the system categorizes the value as high. When the value is between 74 and 50 then it is categorized as medium. If none of the above categories, then it is assigned to the low category.

Implement the above categorization system using java

4. Write a java console application to add two numbers and display the summation.
5. Implement the following GUI application using java swing.

